

## 1.10 Movement Without Drive Power

Note: According to IEC 60204-1 and ISO 13850, emergency devices are not safeguards. They are complimentary protective measures and are not intended to prevent injury.

The risk assessment of the robot application shall conclude if additional emergency stop buttons are need. Emergency stop push-buttons must comply with IEC 60947-5-5 (see section 5.4.2).

### 1.10 Movement Without Drive Power

In the unlikely event of an emergency, you can use **forced back-driving** where you must move robot joint/s, but robot power is either impossible or unwanted.

To perform **forced back-driving** you must push, or pull, the robot arm hard to move the joint. Each joint brake has a friction clutch that enables movement during high forced torque.

Note: In a service situation, the brake on the joints can be released without connected power.



#### WARNING:

- Moving the robot arm manually is intended for urgent emergency purposes only and might damage the robot joints.



## 2 Safety-related Functions and Interfaces

### 2.1 Introduction

Universal Robots e-Series robots are equipped with a range of built-in safety functions as well as safety I/O, digital and analog control signals to or from the electrical interface, to connect to other machines and additional protective devices. Each safety function and I/O is constructed according to EN ISO13849-1:2008 (see chapter 8 for certifications) with Performance Level d (PLd) using a category 3 architecture.

See chapter 13 in part II for configuration of the safety functions, inputs and outputs in the user interface. See chapter 5 for descriptions on how to connect safety devices to I/O.



#### NOTE:

1. The use and configuration of safety functions and interfaces must follow the risk assessment procedures for each robot application. (see chapter 1 section 1.7)
2. If the robot discovers a fault or violation in the safety system (e.g. if one of the wires in the Emergency Stop circuit is cut or a safety limit is violated) then a Stop Category 0 is initiated.
3. The stopping time should be taken into account as part of the application risk assessment



#### DANGER:

1. The use of safety configuration parameters different from those determined by the risk assessment can result in hazards that are not reasonably eliminated or risks that are not sufficiently reduced
2. Ensure tools and grippers are connected appropriately so if there is an interruption of power, no hazards occur!
3. Use caution with 12V, since an error made by the programmer can cause the voltage to change to 24V, which might damage the equipment and cause a fire
4. The end effector is not protected by the UR safety system. The functioning of the end effector and/or connection cable is not monitored

## 2.2 Stop Categories

Depending on the circumstances, the robot can initiate three types of stop categories defined according to IEC 60204-1). These categories are defined in the following table.

| Stop Category | Description                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 0             | Stop the robot by immediate removal of power.                                                                                               |
| 1             | Stop the robot in an orderly, controlled manner. Power is removed once the robot is stopped.                                                |
| 2             | *Stop the robot with power available to the drives, while maintaining the trajectory. Drive power is maintained after the robot is stopped. |

Note: \*Universal Robots robots' Category 2 stops are further described as SS1 or as SS2 type stops according to IEC 61800-5-2.

## 2.3 Safety Functions

Universal Robots robot safety functions, as listed in the table below, are in the robot but are meant to control the robot system i.e. the robot with its attached tool/end effector. The robot safety functions are used to reduce robot system risks determined by the risk assessment. Positions and speeds are relative to the base of the robot.

| Safety Function         | Description                                                                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Joint Position Limit    | Sets upper and lower limits for the allowed joint positions.                                                                                                                                                          |
| Joint Speed Limit       | Sets an upper limit for joint speed.                                                                                                                                                                                  |
| Safety Planes           | Defines planes, in space, that limit robot position. Safety planes limit either the tool/end effector alone or both the tool/end effector and the elbow.                                                              |
| Tool Orientation        | Defines allowable orientation limits for the tool.                                                                                                                                                                    |
| Speed Limit             | Limits maximum robot speed. The speed is limited at the elbow, at the tool/end effector flange, and at the center of the user-defined tool/end effector positions.                                                    |
| Force Limit             | Limits maximum force exerted by the robot tool/end effector and elbow in clamping situations. The force is limited at the tool/end effector, elbow flange and center of the user-defined tool/end effector positions. |
| Momentum Limit          | Limits maximum momentum of the robot.                                                                                                                                                                                 |
| Power Limit             | Limits mechanical work performed by the robot.                                                                                                                                                                        |
| Stopping Time Limit     | Limits maximum time the robot uses for stopping after a protective stop is initiated.                                                                                                                                 |
| Stopping Distance Limit | Limits maximum distance travelled by the robot after a protective stop is initiated.                                                                                                                                  |

## 2.4 Safety Function

The robot also has the following safety inputs:

| Safety Input               | Description                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Emergency Stop Button      | Performs a Stop Category 1 informing other machines using the <i>System Emergency Stop</i> output, if that output is defined.                             |
| Robot Emergency Stop       | Performs a Stop Category 1 via Control Box input, informing other machines using the <i>System Emergency Stop</i> output, if that output is defined.      |
| System Emergency Stop      | Performs a Stop Category 1 on robot only.                                                                                                                 |
| Safeguard Stop             | Performs a Stop Category 2.                                                                                                                               |
| Safeguard Reset            | Returns from the <i>Safeguard Stop</i> state, when an edge on the Safeguard Reset input occurs.                                                           |
| Reduced Mode               | Transitions the safety system to use the <i>Reduced mode</i> limits.                                                                                      |
| 3-Position Enabling Device | Initiates a <i>Safeguard Stop</i> when the enabling device is fully compressed or fully released. When this happens, the enabling device inputs are high. |
| Operational Mode           | Mode to switch, when needed. NOTE: required when a 3-Position Enabling Device is used.                                                                    |

For interfacing with other machines, the robot is equipped with the following safety outputs:

| Safety Output         | Description                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| System Emergency Stop | While this signal is logic low, the <i>Robot Emergency Stop</i> input is logic low or the Emergency Stop button is pressed.                    |
| Robot Moving          | While this signal is logic high, no single joint of the robot moves more than 0.1 rad.                                                         |
| Robot Not Stopping    | Logic high when the robot is stopped or in the process of stopping due to an Emergency Stop or Safeguard Stop. Otherwise it will be logic low. |
| Reduced Mode          | Logic low when the safety system is in Reduced Mode.                                                                                           |
| Not Reduced Mode      | Logic low when the system is not in Reduced Mode.                                                                                              |

All safety I/O are dual channel, meaning they are safe when low (e.g., the Emergency Stop is active when the signals are low).

## 2.4 Safety Function

The safety system acts by monitoring if any of the safety limits are violated or if an Emergency Stop or a Safeguard Stop is initiated.

The reactions of the safety system are:

| Trigger         | Reaction         |
|-----------------|------------------|
| Emergency Stop  | Stop Category 1. |
| Safeguard Stop  | Stop Category 2. |
| Limit Violation | Stop Category 0. |
| Fault Detection | Stop Category 0. |

When performing the application risk assessment, it is necessary to take into account the motion of the robot after a stop has been initiated. In order to ease this process, the safety functions *Stopping Time Limit* and *Stopping Distance Limit* can be used. These safety functions dynamically reduces the speed of the robot motion such that it can always be stopped within the limits. It is important to note that the joint position limits, the safety planes and the tool/end effector orientation limits take the expected stopping distance travel into account i.e. the robot motion will slow down before the limit is reached.

The functional safety can be summarized as:

| Safety Function         | Tolerance | Performance Level | Category |
|-------------------------|-----------|-------------------|----------|
| Emergency Stop          | –         | d                 | 3        |
| Safeguard Stop          | –         | d                 | 3        |
| Joint Position Limit    | 5 °       | d                 | 3        |
| Joint Speed Limit       | 1.15 °/s  | d                 | 3        |
| Safety Planes           | 40 mm     | d                 | 3        |
| Tool Orientation        | 3 °       | d                 | 3        |
| Speed Limit             | 50 mm/s   | d                 | 3        |
| Force Limit             | 25 N      | d                 | 3        |
| Momentum Limit          | 3 kg m/s  | d                 | 3        |
| Power Limit             | 10 W      | d                 | 3        |
| Stopping Time Limit     | 50 ms     | d                 | 3        |
| Stopping Distance Limit | 40 mm     | d                 | 3        |



**WARNING:**

There are two exceptions to the force limiting function that are important when designing an application (Figure 2.1). As the robot stretches out, the knee-joint effect can give high forces in the radial direction (away from the base) at low speeds. Similarly, the short leverage arm, when the tool/end effector is close to the base and moving around the base, can cause high forces at low speeds. Pinching hazards can be avoided by removing obstacles in these areas, placing the robot differently, or by using a combination of safety planes and joint limits to eliminate the hazard by preventing the robot moving into this region of its workspace.

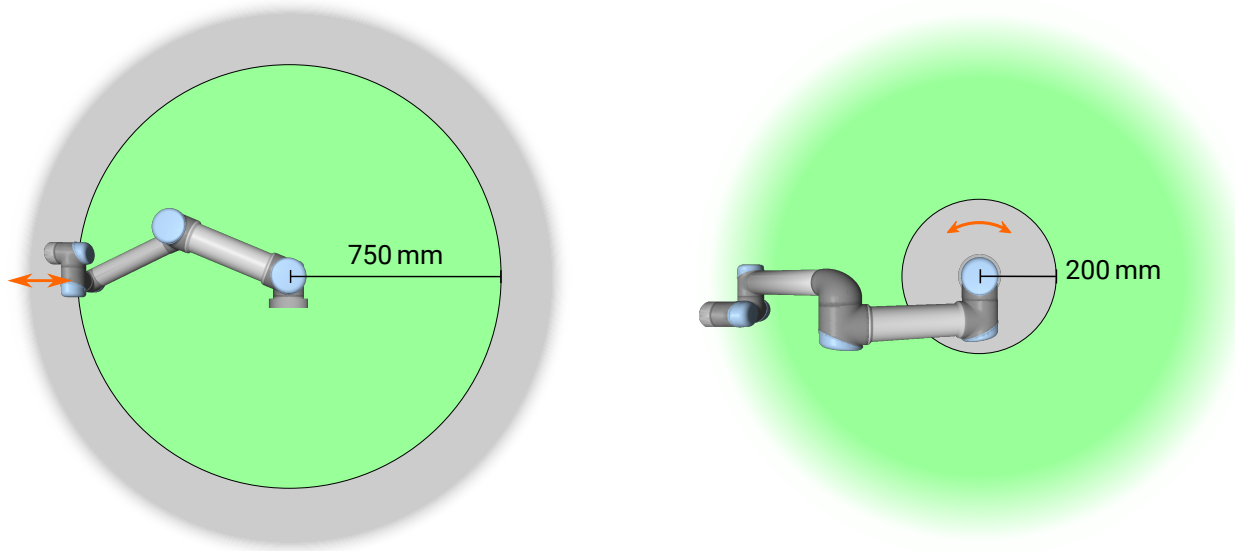


Figure 2.1: Due to the physical properties of the robot arm, certain workspace areas require attention regarding pinching hazards. One area (left) is defined for radial motions when the wrist 1 joint is at least 750 mm from the base of the robot. The other area (right) is within 200 mm of the base of the robot, when moving tangentially.



**WARNING:**

If the robot is used in manual hand-guiding applications with linear movements, the speed limit must be set to maximum 250 mm/s for the tool/end effector and elbow unless a risk assessment shows that higher speeds are acceptable. This will prevent fast movements of the robot elbow near singularities.

## 2.5 Modes

**Normal and Reduced Mode** The safety system has two configurable Modes: **Normal** and **Reduced**. Safety limits can be configured for each of these two modes. Reduced Mode is active when the robot tool/end effector is positioned on the Reduced Mode side of a **Trigger Reduced Mode** plane or when triggered by a safety input.

**Using a plane to trigger Reduced Mode:** When the robot moves from the Reduced Mode side of the trigger plane, back to the Normal Mode side, there is a 20mm area around the trigger plane where both Normal and Reduced Mode limits are allowed. It prevents the Safety Mode from flickering if the robot is right at the limit.

**Using an input to trigger Reduced Mode:** When an input is used (to either start or stop Reduced Mode), up to 500ms can elapse before the new mode limit values are applied. This could happen either when changing Reduced Mode to Normal Mode OR changing Normal Mode to Reduced Mode. It allows the robot to adapt e.g. the speed to the new safety limits.

**Recovery Mode** When a safety limit is violated, the safety system must be restarted. If the system is outside a safety limit at start-up (e.g. outside a joint position limit), the special Recovery Mode is entered. In Recovery Mode, it is not possible to run programs for the robot, but the robot arm can be manually moved back within limits either by using Freedrive Mode or by using the Move tab in PolyScope (see part II PolyScope Manual). The safety limits of Recovery Mode are:

| Safety Function   | Limit     |
|-------------------|-----------|
| Joint Speed Limit | 30 °/s    |
| Speed Limit       | 250 mm/s  |
| Force Limit       | 100 N     |
| Momentum Limit    | 10 kg m/s |
| Power Limit       | 80 W      |

The safety system issues a Stop Category 0 if a violation of these limits appears.



**WARNING:**

Notice that limits for the joint positions, the safety planes, and the tool/end effector orientation are disabled in Recovery Mode. Take caution when moving the robot arm back within the limits.



## 3 Transportation

As supplied on the pallet, the robot and Control Box are a calibrated set. Do not separate them as this would require recalibration.

Only transport the robot in its original packaging. Save the packaging material in a dry place if you want to move the robot later.

When moving the robot from its packaging to the installation space, hold both tubes of the robot arm at the same time. Hold the robot in place until all mounting bolts are securely tightened at the base of the robot.

Lift the Control Box by its handle.



### WARNING:

1. Make sure not to overload your back or other bodyparts when lifting the equipment. Use proper lifting equipment. All regional and national lifting guidelines shall be followed. Universal Robots cannot be held responsible for any damage caused by transportation of the equipment.
2. Make sure to mount the robot according to the instructions in chapter 4.



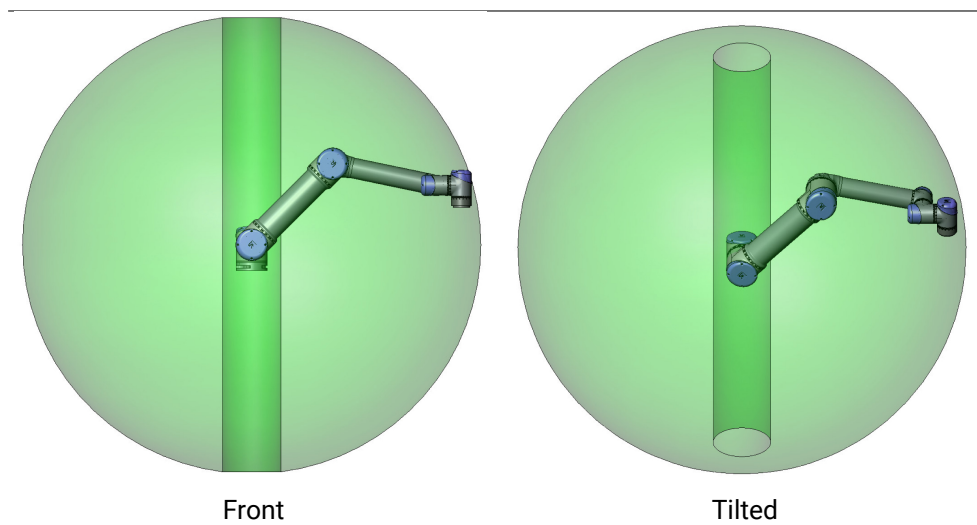
## 4 Mechanical Interface

### 4.1 Introduction

This chapter describes the basics of mounting the parts of the robot system. Electrical installation instructions in chapter 5 must be observed.

### 4.2 Workspace of the Robot

The workspace of the UR5e robot extends 850 mm from the base joint. It is important to consider the cylindrical volume directly above and directly below the robot base when choosing a mounting place for the robot. Moving the tool close to the cylindrical volume should be avoided because it causes the joints to move fast even when the tool is moving slowly, which causes the robot to work inefficiently and makes it difficult to conduct a risk assessment.



### 4.3 Mounting

**Robot Arm** The Robot Arm is mounted using four 8.8 strength, M8 bolts and four 8.5 mm mounting holes at the base. The bolts must be tightened with 20 N m torque.

Use the two Ø8 holes provided, with a pin, to accurately reposition Robot Arm. Note: You can purchase an accurate base counterpart as an accessory. Figure 4.1 shows where to drill holes and mount the screws.

Mount the robot on a sturdy, vibration-less, surface that can withstand at least ten times the full torque of the base joint and at least five times the weight of the Robot Arm. If the robot is mounted on a linear axis, or a moving platform, then the acceleration of the moving mounting base is very low. A high acceleration might cause the robot to make a safety stop.


**DANGER:**

Make sure the Robot Arm is properly and securely bolted in place. Unstable mounting can lead to accidents.


**CAUTION:**

Mount the robot in an environment suited for the IP rating. The robot must not be operated in environments that exceed those corresponding to the IP ratings of the robot (IP54), Teach Pendant (IP54) and Control Box (IP44)

**Tool** The robot tool flange has four M6 thread holes for attaching a tool to the robot. The M6 bolts must be tightened with 8 N m, strength class 8.8. For accurate tool repositioning, use a pin in the Ø6 hole provided. Figure 4.2 shows dimensions and hole pattern of the tool flange. It is recommended to use a radially slotted hole for the pin to avoid over-constraining, while keeping precise position.


**DANGER:**

1. Ensure the tool is properly and securely bolted in place.
2. Ensure the tool is constructed such that it cannot create a hazardous situation by dropping a part unexpectedly.

**Control Box** The Control Box can be hung on a wall or placed on the ground. A clearance of 50 mm on each side of the Control Box is needed for sufficient airflow.

**Teach Pendant** The Teach Pendant can be hung on a wall or on the Control Box. Verify that the cable does not cause tripping hazard.

Note: you can buy extra brackets for mounting the Control Box and Teach Pendant.


**DANGER:**

1. Make sure the Control Box, Teach Pendant and cables do not come into contact with liquids. A wet Control Box could cause fatal injury.
2. Place the Teach Pendant (IP54) and Control Box (IP44) in an environment suited for the IP rating.